

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination
November – December 2012

CL 401 Estimation, Costing & Financial Concepts - I

Date: 04.12.2012, Tuesday

Time: 10:00 p.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Distinguish centre line method to general method of working out quantity of civil work. 03
(b) What are the general rules for taking out quantities and abstract preparation? 03
(c) Draw neat and to the scale foundation excavation plan for foundation of the Figure 1. 03
Calculate quantity of excavation for foundation.

- Q - 2 Calculate following quantity for two room plan shown in Figure 1. Use Centre line method 14
OR Long wall short method.

- 1) B.B.C.C work in foundation (1:3:6)
- 2) Brick work in foundation (1:4)
- 3) Earth filling in foundation
- 4) Brick work in super structure up to 3m height.

OR

- Q - 2 Prepare detail estimates of main items of work for retaining wall shown in Figure 2. 14

Q - 3 Attempt any two

- (a) Calculate quantity of material for 1 m^3 of brick work in CM 1:4 for its rate analysis. Assume suitable data and consider thickness of wall 20 mm. 06
- (b) Derive the rate of 10 cm R.C.C. slab proportion (1:2:4) with 1% steel reinforcement for 10 sq m. 06
- (c) What is overhead cost? How we compensate overhead cost in rate analysis? 06
- (d) State practical application of rate analysis of civil works and parameters which influence it. 06

SECTION - II

- Q - 4 (a) Fill in the blanks 04

1. _____ is use for administrative approval of work.
2. For the changes of some details in existing work _____ estimate is required to prepare.
3. When original sanctioned estimate is exceeded or likely to exceed by more than 5% _____ estimate is prepared.
4. _____ is required for technical sanction of work.

- (b) State rules for deduction in plaster work. 03

- Q - 5 (a) What do you understand with specification of works? Distinguish between brief specifications and detailed specifications. 05

- (b) Draft detail specification for Cement concrete in foundation. (1 : 4 : 8) 05

- (c) Write short note on detail estimate. 04

OR

- Q - 5 (a) Write general specifications of a first class building. 05
 (b) Draft detail specification for plastering cement mortar. 05
 (c) Prepare a preliminary estimate of Figure 1. Take plinth area rate Rs. 1200/- per Sq.m. Assume suitable rates for other works. 04

- Q - 6 (a) What is contract? State types of contract. Distinguish contract to agreement. 05
 (b) Define: Tender, Earnest money, Security deposit, Retention money and competent authority. 05
 (c) State general guidelines for measurement of earth work as per IS: 1200. 04

OR

- Q - 6 (a) What is termination of contract? Draft general conditions for termination of contract. 05
 (b) State different type of tenders. Explain one in detail. 05
 (c) State general guidelines for accurate measurement of brick work as per IS: 1200 04

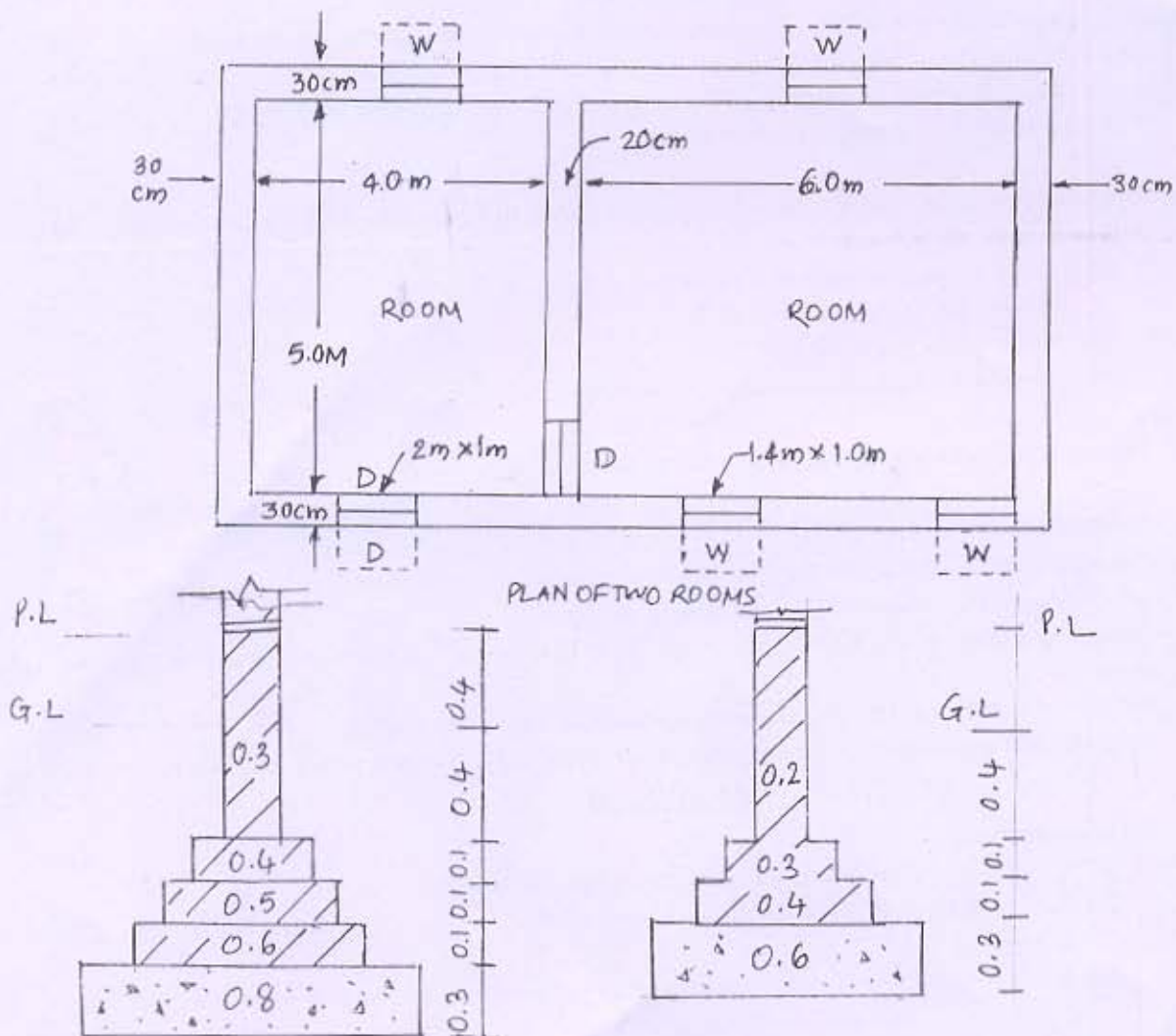
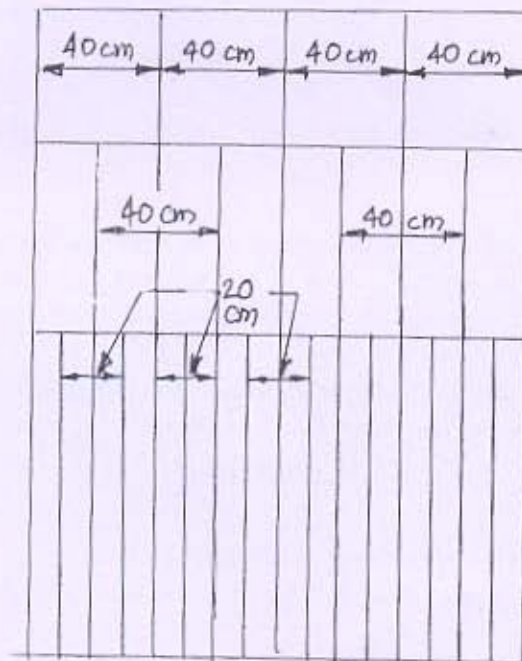
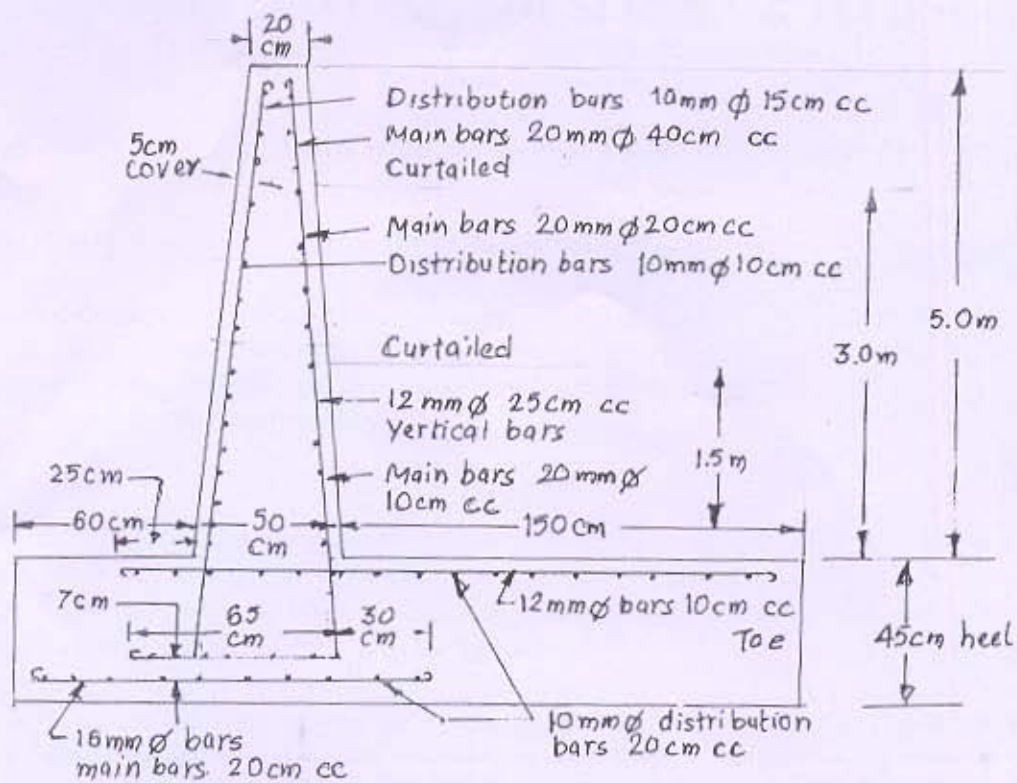


Figure 1. Two room plan with foundation detail



Arrangement of 20 mm ϕ Main Reinforcement Bars

Figure 2: R.C.C retaining wall section
